BTS-V(R)-11-21-0963 Reg. No. | | | | | | | | |

A

B.Tech. Degree V Semester Regular Examination November 2021

ME 19-205-0505 POWER PLANT ENGINEERING

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

(8 × 3 = 24)

I. (a) A power plant has the following annual factors: load factor = 0.75, capacity factor = 0.6, use factor = 0.65. Maximum demand is 60 MW. Estimate the
   (i) Annual energy production
   (ii) Reserve capacity over peak load

(b) Explain the importance of a surge tanks.

(c) Explain the advantages of the process of inter-cooling and reheating in Brayton cycle using p-v and T-s diagram.

(d) Explain the cooling systems in a diesel power plant with a neat sketch.

(e) Distinguish between fire-tube and water-tube boiler.

(f) Explain the functions of
   (i) Moderator
   (ii) Reflectors in nuclear power plant.

(g) Explain thermionic power generation.

(h) Explain the salient features of batch type biogas plant.

PART B

(4 × 12 = 48)

II. A generating station has the following daily load cycle: (12)

Time (hours)	0-6	6-10	10-12	12-16	16-20	20-24
Load (MW)	20	25	30	25	35	20

Draw the load curve and find:
(i) maximum demand (ii) units generated per day (iii) average load
(iv) load factor

OR

III Calculate the cost of generation per kWh for a power station having the following data. (12)

Installed capacity	=	200 MW
Capital cost	=	₹400 Crores
Rate of interest and depreciation	=	12%
Annual cost of fuel salaries and taxation	=	₹5 Crores
Load factor	=	50%

Also estimate the saving in cost per kWh if the annual load factor is raised to 60%

(P.T.O)

IV. (a) Draw the general layout of a diesel engine power plant, explaining its various components. (7)

(b) What is regenerative gas turbine power plant? Explain with a neat sketch. Describe the effectiveness of a re-generator using p-v and T-s diagrams. (5)

OR

V. Air enters the compressor of an air standard Brayton cycle at 0.2 MPa and 12°C. The pressure leaving the compressor is 1.3 MPa, and the maximum temperature in the cycle is 1150°C. If a re-generator with effectiveness 0.9 is incorporated into the cycle, determine (12)

(i) The pressure and temperature at each point in the cycle.

(ii) The compressor work, turbine work, and cycle efficiency.

Assume the isentropic compressor efficiency of 75% and isentropic turbine efficiency of 82%

VI. (a) Describe pressurized water reactor (PWR) with a neat sketch. (6)

(b) Briefly explain the various types of mechanical stokers. (6)

OR

VII. (a) Explain the fluidized bed combustion. (5)

(b) A boiler uses 2250 kg/h of coal. The temperature of the air supplied is 305 K, and the average temperature of the flue gas leaving out of the chimney is 660 K. The 32.5 m high chimney produces a draught of 3 cm of water column. Determine (7)

(i) Quantity of air supplied per kg of coal

(ii) Draught in terms of column if hot gas

(iii) Base diameter of the chimney, assuming that 22% of theoretical draught is used for creating the flow velocity of gases through the chimney.

VIII. (a) What are the various components of a Wind turbine power generation system. Explain. (6)

(b) Explain with sketch the working of an Evacuated Tube Collector. (6)

OR

IX. Describe the working of geothermal energy power plant. Explain the types of geothermal sources. (12)

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